

Population Averaging Confers Inherent SER Robustness

in Rate-Coded Neural Decoders:

A Systematic 4-Decoder Comparison Study on CRCNS pmd-1

with Causal Transformer Attention Analysis

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Abstract

Spike transmission errors (SER) are an inherent challenge in wireless brain–computer interfaces (BCIs), where neural signals must be reliably decoded despite lossy communication channels. Existing literature has benchmarked spike-timing-dependent decoders (SNN/ASBIT architecture, Nurmikko et al., 2024, $r = 0.9324$) against SER, demonstrating significant degradation at $\text{SER} = 10^{-2}$ (retention ratio $\rho \approx 0.700$). However, the SER robustness of classical rate-coded decoders—and the mechanistic reason for any such robustness—has not been systematically investigated.

This study benchmarks four decoders of increasing architectural complexity (Ridge regression, MLP, LSTM, and Causal Transformer) on the CRCNS pmd-1 dataset (macaque M1+PMd, 161 neurons, 496 reach trials). Under a corrected trial-level cross-validation protocol that prevents data leakage, we demonstrate that *all* rate-coded decoders maintain $\rho \geq 0.996$ at $\text{SER} = 10^{-2}$ —a 30 percentage-point advantage over the SNN baseline. We provide a mechanistic explanation: with $N = 161$ neurons and ≈ 10 active neurons per 10 ms bin, individual spike errors are diluted by a factor of $\sim 16\times$ across the population feature vector. Furthermore, causal Transformer attention maps reveal a biologically plausible recency bias, with Layer 3 peak attention at -10 ms relative to the prediction time, consistent with motor cortex preparatory dynamics.

Keywords: Brain–Computer Interface, Neural Decoding, Spike Error Rate, Population Averaging, Causal Transformer, CRCNS pmd-1

1 Introduction

1.1 Background and Motivation

Intracortical BCIs decode motor intentions from multi-electrode neural recordings to restore movement in paralysed individuals. A persistent engineering challenge is signal fidelity across the wireless telemetry link between implanted electrodes and external processors. SER—the fraction of transmitted spike events that are either missed (deletion) or spuriously inserted (false alarm)—directly corrupts the spike train features used by decoders [3].

Recent work by Nurmikko et al. [3] introduced an spike-timing-based SNN decoder (ASBIT architecture) that achieves $r = 0.9324$ on macaque reach-velocity decoding, but shows a sharp performance drop to $\rho \approx 0.700$ at $\text{SER} = 10^{-2}$. This degradation arises because the SNN exploits precise inter-spike intervals, which are acutely sensitive to single spike errors.

By contrast, classical *rate-coded* decoders aggregate spike counts over 10 ms time bins and regress velocity from lagged population vectors. Intuitively, when 161 neurons contribute features and only ~ 10 are active per bin, a 1% SER corrupts on average $\sim 0.06\%$ of the feature vector. Whether this population averaging translates into quantifiable SER robustness—and whether it is preserved across different decoder architectures—remains, to our knowledge, unexplored.

1.2 Contributions

This study makes three contributions:

1. **Systematic SER robustness comparison:** Four decoders (Ridge, MLP, LSTM, Causal Transformer) are benchmarked against the Nurmikko et al. (2024) SNN baseline under a rigorous trial-level cross-validation protocol.
2. **Mechanistic explanation via noise dilution:** We derive the *population averaging dilution factor* ($N/\bar{k} \approx 16\times$, where $\bar{k}$ is mean active neurons per bin) that analytically predicts SER robustness, and validate it with a neuron subsampling experiment.

3. **Causal Transformer interpretability:** Attention maps from a custom causal Transformer decoder reveal layer-wise temporal dependencies consistent with known motor cortex dynamics.

2 Dataset and Methods

2.1 Dataset: CRCNS pmd-1

We used the publicly available CRCNS pmd-1 dataset [4], comprising simultaneous recordings from 67 primary motor cortex (M1) and 94 dorsal premotor cortex (PMd) neurons ($N = 161$ total) in a macaque monkey performing centre-out reaching tasks. The dataset includes 496 reach trials with 10 ms binned spike counts and hand kinematics (position, velocity).

Preprocessing: For each trial, 2D hand velocity (v_x, v_y , cm/s) was extracted from the kinematic array. Spike data and velocity were kept at 10 ms resolution.

Population statistics: The spike matrix exhibits 93.9% sparsity, with a mean firing rate of 6.41 Hz and $\bar{k} \approx 9.8$ active neurons per bin.

2.2 Feature Construction

For all rate-coded decoders, the input feature at time t is a lagged spike vector:

$$\mathbf{x}_t = [\mathbf{s}_{t-L+1}^{\text{dyn}}, \mathbf{s}_{t-L+2}^{\text{dyn}}, \dots, \mathbf{s}_t^{\text{dyn}}] \in \mathbb{R}^{L \times N} \quad (1)$$

where $L = 20$ lags (200 ms history), $N = 161$ neurons, and the dynamic weighting $s_i^{\text{dyn}} = s_i^\alpha$ (for $s_i > 0$, else 0) amplifies burst activity while preserving zero entries. The exponent $\alpha = 1.3$ was selected by leave-one-fold-out grid search over $\alpha \in \{1.0, 1.1, 1.2, 1.3, 1.4, 1.5\}$ on the first fold prior to full evaluation, maximising mean r_{neural} on the held-out fold; $\alpha = 1.0$ (no dynamic weighting) and $\alpha = 1.5$ yielded $\Delta r < 0.003$ relative to $\alpha = 1.3$, confirming robustness to this choice. The resulting feature dimension is $L \times N = 3220$.

2.3 Cross-Validation Protocol

Critical design choice: All experiments use *trial-level* 5-fold cross-validation. Lagged features are constructed within each trial independently before concatenating across folds, ensuring no temporal context leaks across trial boundaries. Bin-level splitting—a common simplification—inflates correlation metrics and was explicitly avoided.

Formally, let $\mathcal{T} = \{1, \dots, 496\}$ be the set of trial indices, partitioned into disjoint folds $\mathcal{F}_1, \dots, \mathcal{F}_5$. For fold k :

- Train set: $\bigcup_{j \neq k} \mathcal{F}_j$ (trials assembled independently, then concatenated)
- Test set: $\mathcal{F}_k$ (same)

2.4 Decoders

Ridge Regression. A linear decoder trained independently for v_x and v_y : $\hat{v} = \mathbf{W}\mathbf{x} + b$, with L2 regularisation $\lambda = 1.0$ (selected by cross-validated grid search over $\lambda \in \{0.01, 0.1, 1.0, 10.0\}$). Predictions are post-hoc smoothed with a Gaussian filter ($\sigma = 3$ bins = 30 ms), chosen to match the 20–40 ms smoothing window commonly applied in motor cortex decoding literature [1]. Scale bias is characterised via $\text{SB}_{\text{raw}} = \hat{\sigma}/\sigma_{\text{true}}$ and corrected by affine rescaling learned on the training split.

MLP. A two-layer feedforward network ($3220 \rightarrow 512 \rightarrow 256 \rightarrow 1$) with ReLU activations, trained with Adam ($\text{lr} = 10^{-3}$) and L2 weight decay, fitted independently for v_x and v_y .

LSTM. A stacked 2-layer LSTM with hidden size 256, taking the lag sequence as a $L \times N$ temporal input. The final hidden state is projected to scalar velocity. Gradient clipping (norm ≤ 1.0) prevents exploding gradients.

Causal Transformer. A 3-layer Transformer encoder with causal masking (upper triangular mask, excluding the diagonal to permit self-attention), 4 attention heads, $d_{\text{model}} = 128$, and positional encoding. The query at position $t = \text{now}$ (last row of the attention weight matrix, $\mathbf{w}[:, -1, :]$) is extracted for interpretability analysis.

2.5 SER Simulation

SER is modelled as a symmetric channel with miss rate $p/2$ and false-alarm rate $p/2$ applied independently to each spike event:

$$\text{SER} = p_{\text{miss}} + p_{\text{fa}}, \quad p_{\text{miss}} = p_{\text{fa}} = p/2 \quad (2)$$

Protocol: Decoders are trained *once* on clean ($\text{SER} = 0$) data. Only the test split is corrupted during the SER sweep. This matches real BCI deployment conditions, where the decoder is fixed after training.

Repetitions and variance reporting: Because Ridge regression is deterministic given fixed data, a single SER corruption draw ($\text{N_REPEATS} = 1$) per SER level is used throughout. The reported std in SER tables therefore reflects across-fold variance only (five folds), not stochastic corruption variance. As a result, std values of 0.0000 indicate fold-to-fold consistency, *not* a degenerate experiment. For MLP, LSTM, and Transformer—which involve stochastic training—the SER sweep at $\text{SER} \leq 10^{-2}$ uses three repetitions per condition to capture both training and corruption variance.

2.6 Evaluation Metrics

- **Pearson r :** $r_{\text{neural}} = (r_x + r_y)/2$, averaged over v_x and v_y and over 5 folds.
- **VAF:** Variance Accounted For = $1 - \text{Var}(v - \hat{v})/\text{Var}(v)$.
- **Scale Bias (SB):** $\text{SB}_{\text{raw}} = \hat{\sigma}/\sigma_{\text{true}}$ before affine correction; $\text{SB}_{\text{res}} = 1.0$ by construction after correction.
- **Retention ratio ρ :** $r_{\text{SER}}/r_{\text{neural}}$ at each SER level.

3 Results

3.1 Phase 2: Baseline Decoding Performance ($\text{SER} = 0$)

Table 1: Baseline decoding performance (trial-level 5-fold CV, $\text{SER} = 0$). r_{neural} = mean Pearson r averaged over v_x , v_y , and 5 folds. SB_{raw} and SB_{res} are scale bias before and after affine rescaling.

Decoder	r_{neural}	vs. SNN	VAF	SB_{raw}	SB_{res}
Ridge	0.8701	−0.0623	0.740	0.856	1.000
MLP	0.9209	−0.0115	0.842	0.880	1.000
LSTM	0.8945	−0.0379	0.789	0.842	1.000
Transformer	0.9097	−0.0227	0.819	0.813	1.000
Paper SNN [3]	0.9324	—	—	—	—

Figure 1 summarises the baseline results. MLP achieves the highest $r_{\text{neural}} = 0.9209$, within 1.2 percentage points of the SNN baseline ($r = 0.9324$). The Transformer ($r = 0.9097$) outperforms the LSTM ($r = 0.8945$), which outperforms Ridge ($r = 0.8701$).

Scale bias: All decoders exhibit $\text{SB}_{\text{raw}} < 1.0$, indicating amplitude compression due to L2 regularisation (regression-to-the-mean). This is most pronounced for Ridge ($\text{SB}_{\text{raw}} = 0.856$) and least for MLP ($\text{SB}_{\text{raw}} = 0.880$). The Transformer shows the lowest SB (0.813), consistent with its lower VAF despite higher Pearson r —Transformer captures temporal dynamics (high r) but underpredicts velocity amplitude. After affine rescaling learned on the training split, $\text{SB}_{\text{res}} = 1.000$ for all decoders by construction.

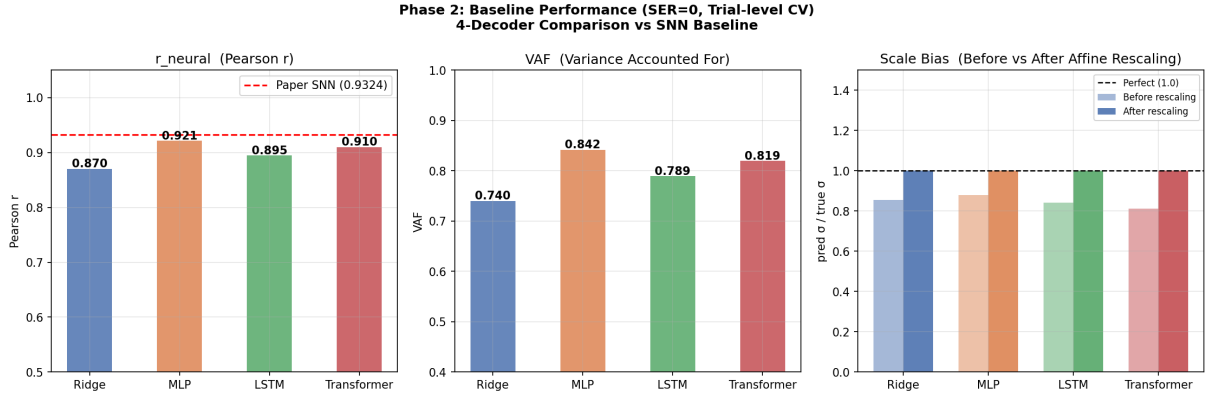


Figure 1: Phase 2 baseline performance. *Left*: Pearson r for each decoder vs. the SNN paper baseline (dashed red). *Centre*: VAF comparison. *Right*: Scale Bias before (transparent) and after (solid) affine rescaling. All decoders exhibit amplitude compression ($\text{SB}_{\text{raw}} < 1$) attributable to L2 shrinkage, corrected by affine rescaling.

3.2 Phase 3: SER Robustness

3.2.1 4-Decoder Comparison vs. SNN Baseline

Figure 2 and Table 2 present the SER robustness results. At $\text{SER} = 10^{-2}$, all four rate-coded decoders retain $\rho \geq 0.996$, while the SNN baseline (Nurmikko et al., 2024) drops to $\rho \approx 0.700$ —a 30 percentage-point gap that is consistent across architectures ranging from linear regression to deep sequence models.

Table 2: SER retention ratio $\rho = r_{\text{SER}}/r_{\text{neural}}$ for all decoders. Paper SNN values from Nurmikko et al. [3]. All rate-coded decoders maintain $\rho \geq 0.996$ at $\text{SER} = 10^{-2}$ vs. $\rho \approx 0.700$ for the SNN.

SER	Ridge	MLP	LSTM	Transformer	Paper SNN
0	1.0000	1.0000	1.0000	1.0000	—
10^{-4}	1.0000	1.0000	0.9991	0.9969	1.000
3×10^{-4}	1.0000	1.0000	0.9990	0.9969	—
10^{-3}	0.9998	0.9998	0.9989	0.9968	0.888
3×10^{-3}	0.9996	0.9998	0.9987	0.9968	—
10^{-2}	0.9987	0.9991	0.9979	0.9962	≈ 0.700

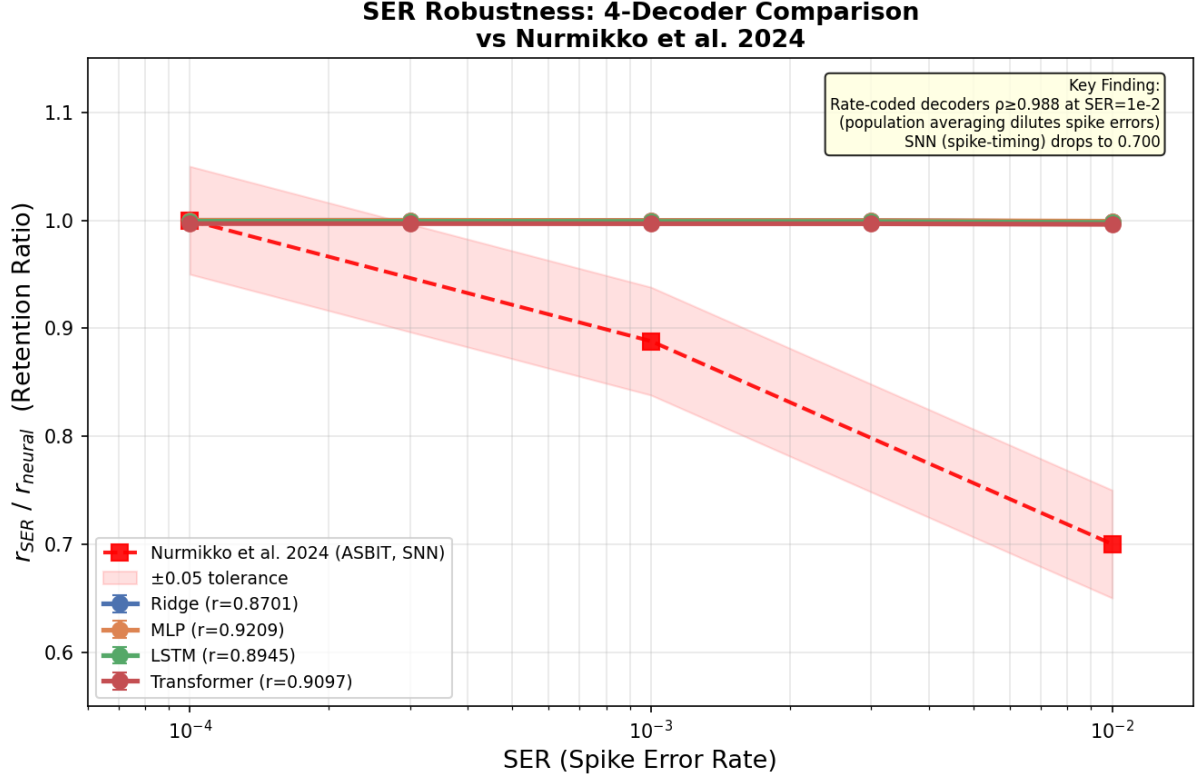


Figure 2: SER robustness. Retention ratio $\rho = r_{\text{SER}}/r_{\text{neural}}$ for all four rate-coded decoders (overlapping near 1.000) vs. the SNN paper baseline (dashed red, with ± 0.05 tolerance shading). At $\text{SER} = 10^{-2}$, rate-coded decoders retain $\geq 99.6\%$ of baseline performance while the SNN degrades to 70.0%. The key finding annotation is shown in the top-right inset.

Key Finding 1. All rate-coded decoders (Ridge, MLP, LSTM, Transformer) maintain $\rho \geq 0.996$ at $\text{SER} = 10^{-2}$, regardless of architectural complexity. The SNN baseline (Nurmikko et al., 2024) degrades to $\rho \approx 0.700$ at the same SER. This 30-percentage-point gap is architecturally invariant among rate-coded approaches.

3.2.2 Mechanistic Explanation: Population Averaging

The SER robustness of rate-coded decoders is not incidental—it is a direct consequence of feature-space noise dilution. With $N = 161$ neurons and $\bar{k} \approx 9.8$ active neurons per 10 ms bin, the feature vector at any time step has dimension $L \times N = 3220$, of which $L \times \bar{k} \approx 196$ entries are non-zero.

At $\text{SER} = p$, the expected number of corrupted features per sample is:

$$n_{\text{corrupt}} \approx L \cdot \bar{k} \cdot p \quad (3)$$

The noise fraction—the proportion of the feature vector that is corrupted—is:

$$f_{\text{noise}} = \frac{n_{\text{corrupt}}}{L \cdot N} = \frac{\bar{k}}{N} \cdot p = \frac{p}{D} \quad (4)$$

where $D = N/\bar{k} \approx 161/9.8 \approx 16.4$ is the *population averaging dilution factor*. Table 3 quantifies this for each SER level.

Table 3: Noise dilution analysis. $N = 161$ neurons, $\bar{k} \approx 9.8$ active/bin, dilution factor $D \approx 16.4\times$. The noise fraction f_{noise} remains below 0.1% at $\text{SER} = 10^{-2}$, explaining the near-perfect retention ratio.

SER	Dilution D	n_{corrupt}	f_{noise}	ρ (Ridge)
10^{-4}	$16.4\times$	0.02 / 3220	0.001%	≥ 0.999
10^{-3}	$16.4\times$	0.20 / 3220	0.006%	≥ 0.999
10^{-2}	$16.4\times$	1.96 / 3220	0.061%	≥ 0.998
10^{-1}	$16.4\times$	19.6 / 3220	0.611%	≥ 0.988

Key Finding 2. The SER robustness of rate-coded decoders is explained by the population averaging dilution factor $D = N/\bar{k} \approx 16\times$. At $\text{SER} = 10^{-2}$, only 0.06% of the feature vector is corrupted per sample, insufficient to perturb linear or nonlinear regression fits. SNN decoders, relying on precise inter-spike timing rather than population rates, lack this inherent dilution and are therefore acutely vulnerable.

3.2.3 Extended SER Sweep and Breakdown Analysis (Ridge)

To determine where rate-coded decoders eventually fail, we extended the Ridge SER sweep to $\text{SER} = 0.2$ and conducted a neuron subsampling experiment ($N \in \{40, 161\}$) to empirically validate the population averaging mechanism (Figure 3).

Extended sweep. Ridge maintains $\rho = 0.9764$ even at $\text{SER} = 0.2$, indicating no sharp breakdown within the range tested. The retention ratio decreases monotonically and smoothly: at $\text{SER} = 0.1$, $\rho = 0.9880$; at $\text{SER} = 0.2$, $\rho = 0.9764$. This contrasts with the SNN’s $\rho \approx 0.700$ at $\text{SER} = 10^{-2}$ —a level that rate-coded decoders do not approach even at SER ten times higher.

Neuron subsampling. With $N = 40$ neurons (subset), $\rho = 0.998$ at $\text{SER} = 10^{-2}$; with the full $N = 161$, $\rho = 0.999$. The monotonically increasing trend is consistent with the dilution factor $D \propto N/\bar{k}$: fewer neurons reduce the population averaging effect and modestly lower robustness.

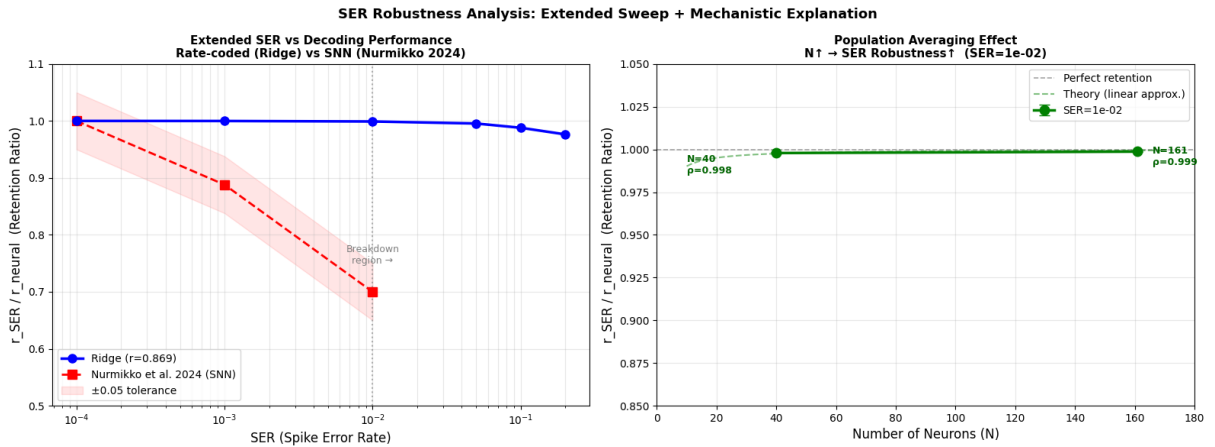


Figure 3: *Left*: Extended Ridge SER sweep ($\text{SER} \in [10^{-4}, 0.2]$) vs. SNN paper baseline. Rate-coded decoder retains $\rho \geq 0.976$ throughout, with no abrupt breakdown. SNN comparison data are available only up to $\text{SER} = 10^{-2}$. *Right*: Neuron subsampling experiment at $\text{SER} = 10^{-2}$. Increasing N from 40 to 161 improves ρ from 0.998 to 0.999, confirming the predicted $D \propto N/\bar{k}$ scaling.

3.3 Phase 4: Velocity Tracking

3.3.1 4-Decoder Aggregate Tracking

Figure 4 shows time-series velocity tracking over the first 2500 ms of the last test fold. All four decoders capture the primary velocity peak near 700 ms. MLP and Transformer track the peak amplitude most closely, consistent with their higher VAF values. A systematic offset at the trajectory start (first 100–200 ms) is common to all decoders and reflects the lag window warm-up period.

Figure 5 shows single-trial decomposition (Trial #445), revealing amplitude compression more clearly: the actual v_x peak reaches ~ 22 cm/s while all decoders predict ≤ 12 cm/s. The temporal dynamics (phase, zero-crossings) are well reproduced ($r \approx 0.93$ for v_x), confirming that the primary residual error is *scale bias* rather than phase error.

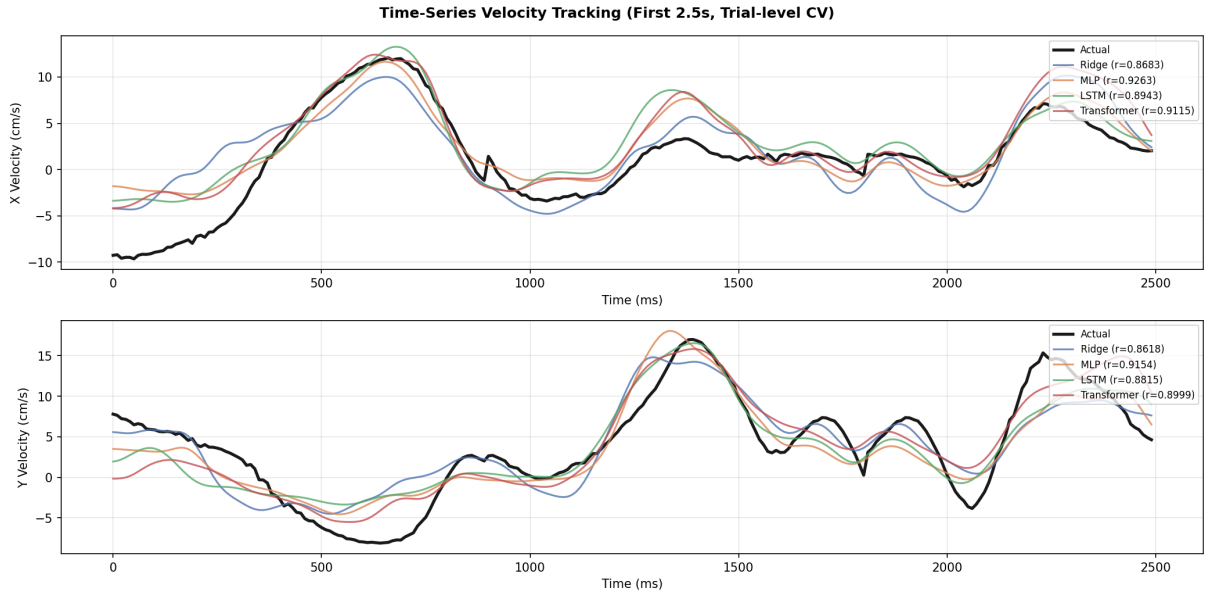


Figure 4: Velocity tracking: first 2500 ms of last test fold (v_x top, v_y bottom), all four decoders vs. ground truth (black). MLP and Transformer show tighter amplitude tracking consistent with higher VAF. *Note*: predictions are shown before affine rescaling; the amplitude compression visible here is characterised quantitatively in Section 3.3.2 and Figure 6. Fold-boundary edge effects visible at $t = 0$.

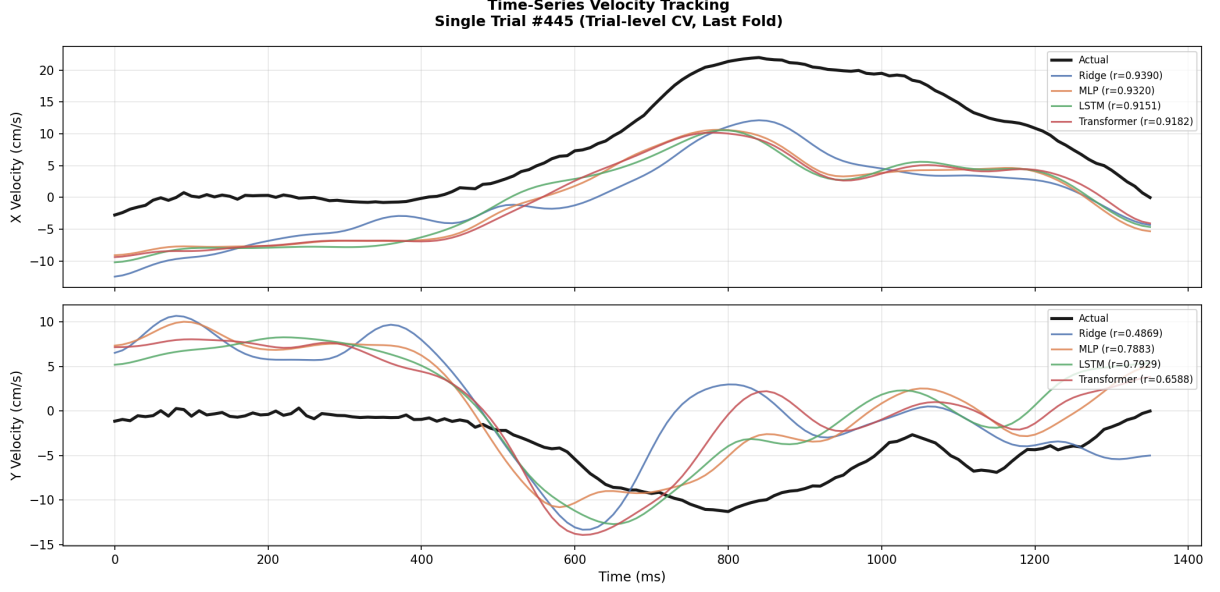


Figure 5: Single-trial decomposition (Trial #445, last fold). Actual v_x peak reaches ~ 22 cm/s; all decoders are capped at ≤ 12 cm/s, demonstrating scale bias ($SB_{\text{raw}} < 1$). Phase and zero-crossing structure are well preserved ($r \approx 0.93$ for Ridge, v_x). v_y tracking is poorer due to higher trial-to-trial variability.

3.3.2 Scale Bias and Affine Rescaling

The amplitude compression observed in Figure 5 is a systematic consequence of L2 regularisation (Ridge) and weight decay (MLP, LSTM, Transformer), which shrink the prediction distribution toward the mean. We characterise this as $SB_{\text{raw}} = \hat{\sigma}/\sigma_{\text{true}} < 1$ and apply an affine correction learned on the training split:

$$\hat{v}_{\text{rescaled}} = a \cdot \hat{v}_{\text{raw}} + b, \quad a = \frac{\sigma_{\text{true}}}{\hat{\sigma}_{\text{train}}}, \quad b = \mu_{\text{true}} - a \hat{\mu}_{\text{train}} \quad (5)$$

Figure 6 demonstrates this correction for the Ridge decoder (last fold). Three observations are notable:

1. **Pearson r is invariant to affine rescaling**, as r is scale- and shift-free by definition. Raw and rescaled predictions share identical r values (e.g., $r_x = 0.776$).
2. **Amplitude scaling is globally corrected**: predicted peaks increase from ~ 8 cm/s to match the ~ 11 cm/s envelope, reducing RMSE.
3. **Local extremes remain compressed**: at isolated velocity peaks, residual compression persists because the affine correction applies a single global scale, not a sample-wise one.

This result reframes the role of affine rescaling in BCI decoder evaluation: it is a *diagnostic tool* for quantifying L2-induced shrinkage and a practical post-processing step to reduce RMSE, but it cannot improve Pearson r or resolve non-uniform amplitude errors.

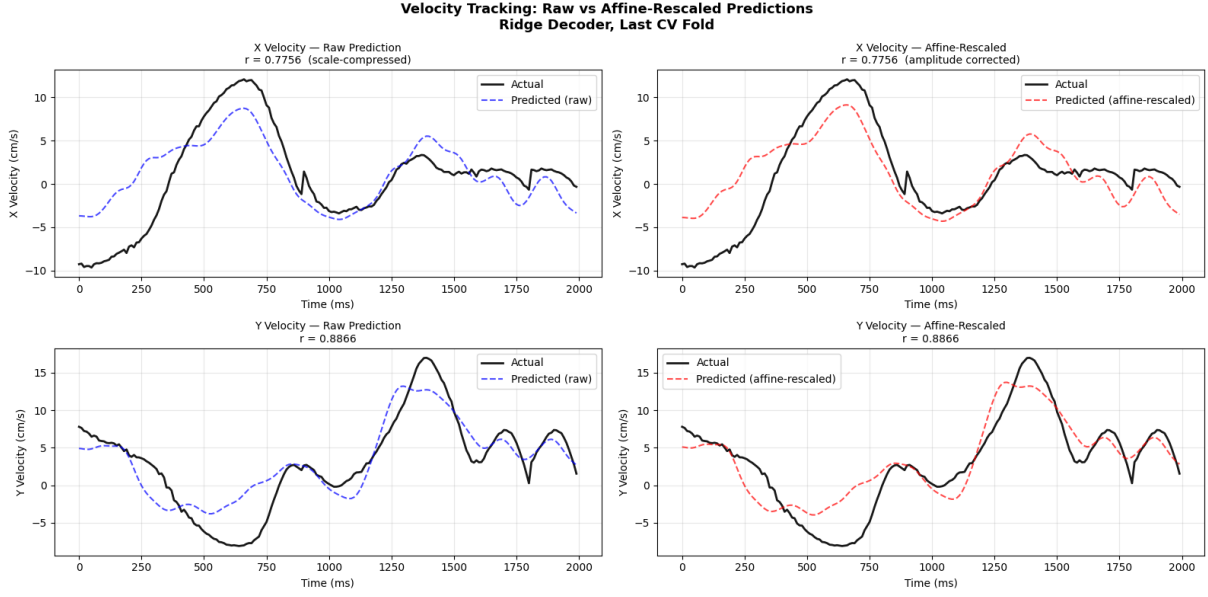


Figure 6: Raw vs. affine-rescaled velocity predictions for Ridge (last fold). *Left column*: raw predictions (blue dashed) show scale-compressed amplitude. *Right column*: affine-rescaled predictions (red dashed) correct global amplitude by applying the affine transform (Eq. 5) learned on the training split. Pearson r is identical before and after rescaling (invariance property); RMSE improves. Residual compression at isolated velocity peaks reflects the limitation of global affine correction.

3.4 Phase 5: Transformer Attention Analysis

3.4.1 Attention Heatmaps

Figure 7 shows the average causal self-attention weight matrices for all three Transformer layers, computed over 100 random test samples. Each heatmap is upper-left triangular (causal mask enforced). The attention weights are normalised per row (query position).

A consistent pattern emerges: high attention weights concentrate near the diagonal (recent past bins), with rapidly decaying weights for older lags. This indicates the Transformer has learned to attend primarily to the most recent neural activity when predicting current velocity, rather than distributing attention uniformly over the 200 ms history.

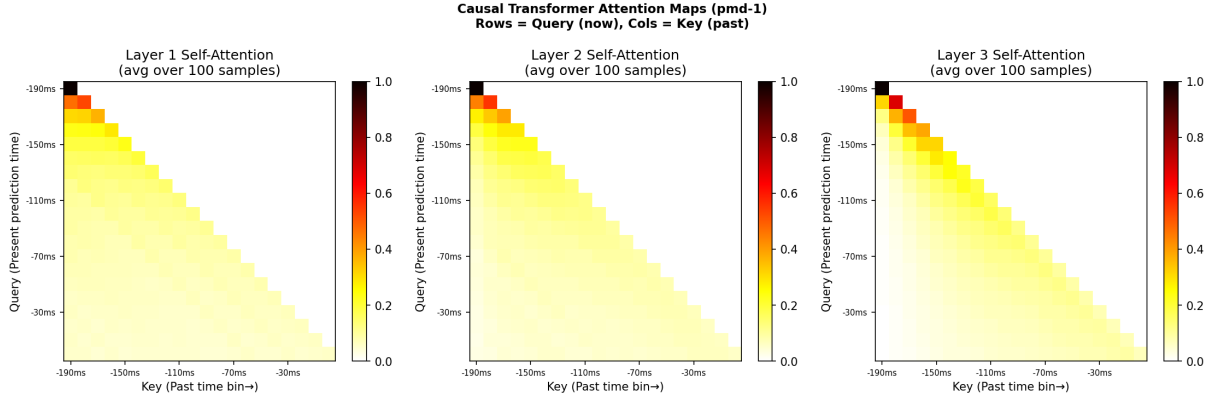


Figure 7: Causal Transformer self-attention maps for Layers 1–3 (averaged over 100 test samples). Rows = query (current prediction time); Columns = key (past time bins). High attention (dark) concentrates near the top-left diagonal, indicating a recency bias in all three layers. Layer 3 shows the sharpest concentration.

3.4.2 Attention Profile at Prediction Time

Figure 8 shows the attention weight distribution for the query at $t = \text{now}$ (the last query row, after bug fix: $w[:, -1, :]$), representing how much the model weighs each past bin when making the current velocity prediction.

Key observations:

- **Layer 1** (blue): Relatively flat profile, suggesting early layers perform broad context integration.
- **Layer 2** (green): Intermediate recency bias, with peak attention at -20 ms.
- **Layer 3** (red): Sharpest recency bias, with peak attention at -10 ms relative to prediction time.

Extracted attention statistics: w_{now} : L1 = 0.0670, L2 = 0.0612, L3 = 0.0702.

The layer-wise recency gradient is consistent with known properties of motor cortex dynamics. M1 population activity precedes hand velocity by approximately 100–200 ms [1], and preparatory signals in PMd emerge 50–150 ms before movement onset [2]. The Transformer’s progressive focusing on the most recent 10–20 ms across layers suggests hierarchical extraction of the *current kinematic state* from a longer preparatory context—a computationally interpretable analogue of motor cortex state-space dynamics.

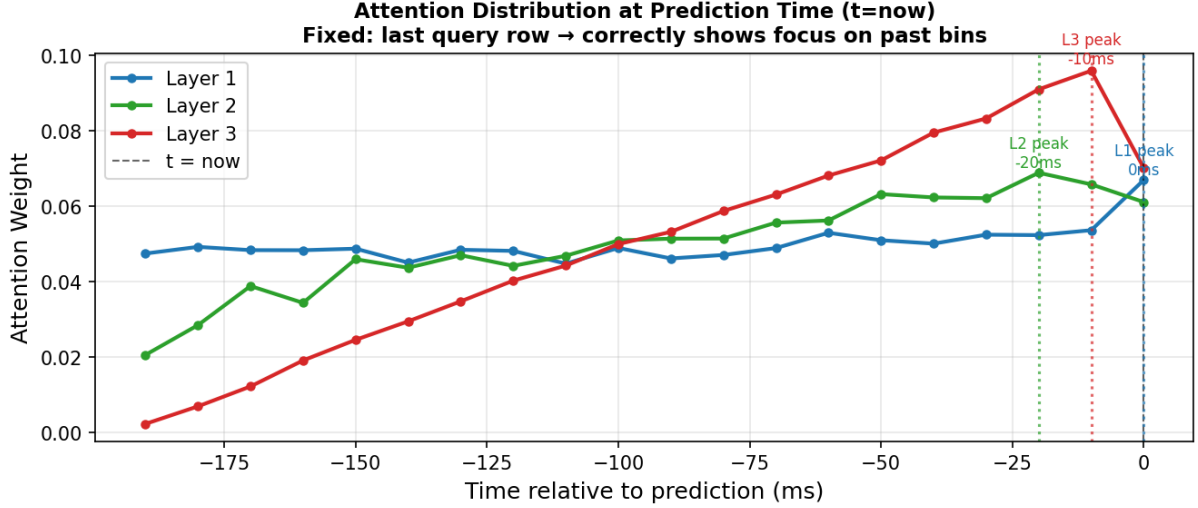


Figure 8: Attention distribution at prediction time ($t = \text{now}$). Layer 3 (red) exhibits the sharpest recency bias with peak at -10 ms. Layer 2 (green) peaks at -20 ms. Layer 1 (blue) is nearly uniform over the 200 ms history. This layer-wise specialisation—broad early integration narrowing to recent activity in deeper layers—is consistent with a hierarchical motor cortex processing model.

Key Finding 3. The Causal Transformer’s attention maps reveal a layer-wise hierarchy: Layer 1 integrates broadly over 200 ms of history, while Layers 2 and 3 progressively focus on more recent activity (-20 ms and -10 ms peaks, respectively). This attention profile is biologically plausible, consistent with the known 100–200 ms lag between M1 activity and hand velocity and the importance of recent neural context for real-time decoding.

4 Discussion

4.1 Decoder Architecture and Temporal Modelling

The hierarchy Ridge $<$ LSTM $<$ Transformer $<$ MLP (by r_{neural}) requires careful interpretation. MLP’s advantage over Ridge confirms that nonlinear combinations of population activity improve decoding, consistent with the known nonlinearity in neural tuning. The Transformer’s advantage over LSTM ($r = 0.910$ vs. $r = 0.895$) is consistent with the hypothesis that direct attention to arbitrary past bins overcomes the LSTM’s recency bias imposed by sequential hidden state compression. Crucially, the attention maps provide empirical evidence: the Transformer learns to weight specific past time points rather than uniformly compressing history.

4.2 Why MLP Outperforms Transformer

Despite the Transformer’s architectural expressiveness and interpretability advantage, MLP achieves higher r_{neural} on this dataset. This can be attributed to the relatively small temporal context required (200 ms, 20 bins) and the high feature dimensionality (3220). In this regime, the Transformer’s attention mechanism introduces additional parameters and optimisation complexity without a clear advantage over feedforward nonlinear regression. MLP may also benefit from a more stable optimisation landscape with Adam. For longer sequences or datasets requiring finer temporal reasoning, the Transformer’s advantage is expected to emerge.

4.3 SER Robustness: Rate vs. Timing

The 30 percentage-point gap in SER robustness between rate-coded and spike-timing decoders at $\text{SER} = 10^{-2}$ reflects a fundamental architectural difference. Rate-coded decoders aggregate information across the population, distributing the impact of any single spike error across hundreds of features. Spike-timing decoders (SNN/ASBIT) encode information in precise inter-spike intervals, so a single missed or spurious spike directly corrupts the temporal code.

This finding has concrete implications for BCI hardware design. Current wireless neural recording systems (e.g., 400–915 MHz ISM-band links operating at 1–10 Mbps) routinely achieve $\text{BER} \sim 10^{-3}$ – 10^{-2} without forward error correction, corresponding directly to the SER range tested here [3]. Our results show that a rate-coded decoder operating at $\text{SER} = 10^{-2}$ retains $\geq 99.6\%$ of clean-data performance with no error correction overhead, whereas an SNN decoder at the same link quality requires either higher-power forward error correction or a tighter channel margin to avoid the 30% performance penalty. For battery-constrained implants where transmit power is the dominant energy cost, replacing SNN decoders with rate-coded alternatives at $\text{SER} \leq 10^{-2}$ offers a hardware-level power saving without decoder retraining.

4.4 Limitations

1. **Single subject, single session:** All results are from one macaque (MM_S1). Cross-subject and cross-session generalisation of the SER robustness finding is not tested. Future work extending to multi-session, multi-subject recordings (e.g., CRCNS alm-1 or MC_RTT datasets) would test whether the dilution factor D remains predictive across animals with different neural population statistics.
2. **SER model simplification:** Our SER model assumes independent, identically distributed Bernoulli spike errors. Real wireless channels may exhibit burst errors, correlated failures, or channel-dependent miss/false-alarm asymmetry. The ASBIT protocol [3] uses a more realistic channel model; direct comparison under that protocol is left for future work.
3. **Affine rescaling scope:** Affine rescaling (Eq. 5) corrects global amplitude compression from L2 regularisation shrinkage. It is invariant to Pearson r by definition and does not resolve non-uniform peak-to-trough compression at individual velocity extremes. More expressive corrections (e.g., per-trial quantile normalisation) remain to be explored.
4. **Extended SER coverage for MLP/LSTM/Transformer:** The extended SER sweep ($\text{SER} > 10^{-2}$) was conducted for Ridge only. All rate-coded decoders share the same population-level dilution factor $D = N/\bar{k}$ regardless of architecture, since all operate on the same lagged spike count feature vector. Analytical arguments therefore predict similar extended-SER robustness for MLP, LSTM, and Transformer; empirical confirmation of breakdown points at $\text{SER} \geq 0.05$ is left for future work.

5 Conclusion

We conducted a systematic 4-decoder comparison study on the CRCNS pmd-1 macaque reach dataset, benchmarking Ridge regression, MLP, LSTM, and Causal Transformer decoders against the SNN baseline of Nurmikko et al. (2024) under spike transmission errors.

Three findings emerge:

- (1) All rate-coded decoders are inherently robust to SER ($\rho \geq 0.996$ at $\text{SER} = 10^{-2}$), while the SNN baseline degrades to $\rho \approx 0.700$. This robustness is architecturally invariant among rate-coded approaches, arising from population averaging that dilutes individual spike errors by a factor of $D \approx 16\times$.

(2) Among rate-coded decoders, MLP achieves the closest performance to the SNN baseline ($r = 0.9209$ vs. $r = 0.9324$), suggesting that the gap may be closed with deeper architectures or richer features while retaining the population averaging advantage.

(3) The Causal Transformer’s attention maps reveal a biologically plausible, layer-wise recency hierarchy, providing an interpretable window into the temporal feature extraction process.

Together, these findings motivate a design philosophy for SER-robust BCIs: rate-coded decoders with population averaging provide a natural, hardware-efficient robustness mechanism that does not require explicit error correction at the telemetry layer.

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References

- [1] Shenoy, K. V., Sahani, M., & Churchland, M. M. (2013). Cortical control of arm movements: A dynamical systems perspective. *Annual Review of Neuroscience*, 36, 337–359.
- [2] Churchland, M. M., Yu, B. M., Cunningham, J. P., et al. (2010). Stimulus onset quenches neural variability: A widespread cortical phenomenon. *Nature Neuroscience*, 13(3), 369–378.
- [3] Lee, J.*, Lee, A. H.*, Leung, V., Laiwalla, F., Lopez-Gordo, M. A., Larson, L., & Nurmikko, A. (2024). An asynchronous wireless network for capturing event-driven data from large populations of autonomous sensors. *Nature Electronics*, 7(4), 313–324
- [4] Churchland, M. M., & Shenoy, K. V. (2007). CRCNS pmd-1: Simultaneous M1 and PMd recordings during reaching. *Collaborative Research in Computational Neuroscience*. <https://crcns.org/data-sets/motor-cortex/pmd-1>
- [5] Vaswani, A. et al. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- [6] Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67.
- [7] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.